TaxSaathi

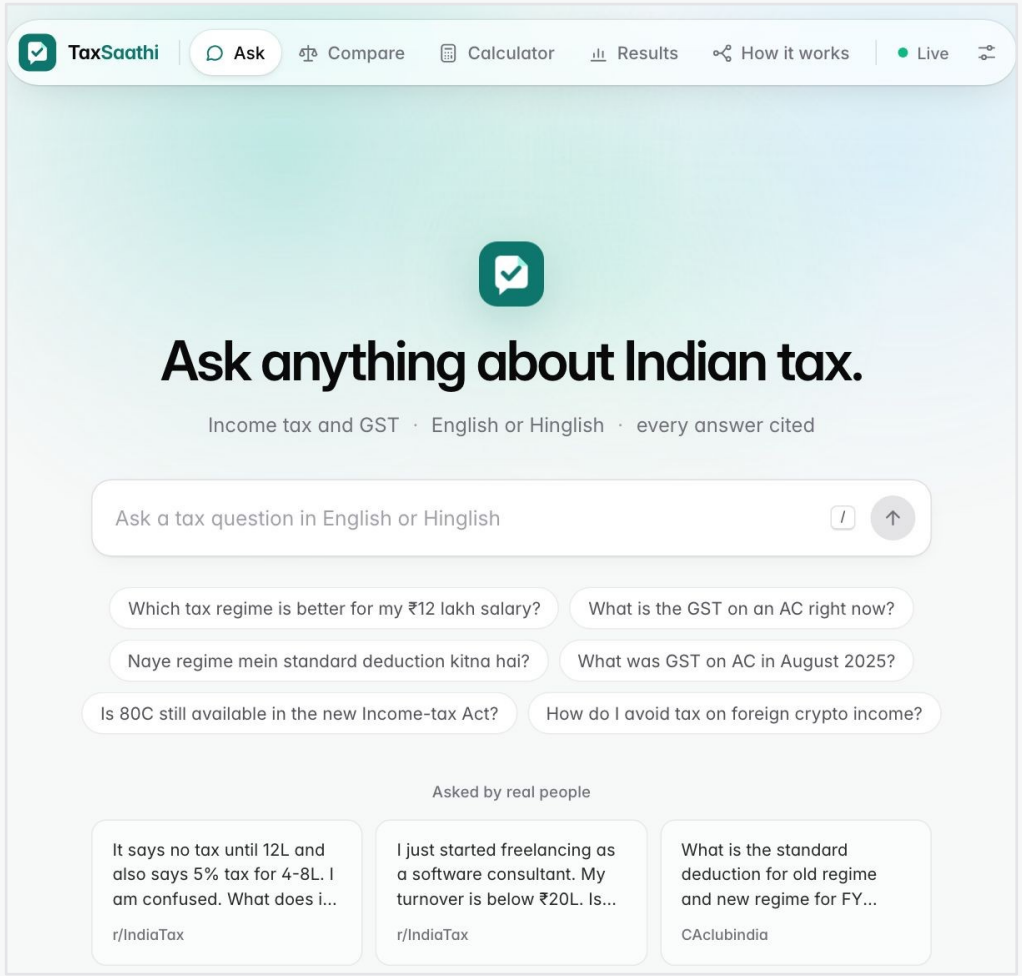
Ask an Indian tax question. Get a short answer that cites the official rule, for the right date.

Speech bubble
your companion

Folded corner
an official source

Check
verified

4th-year NLP course project · Team of 4 · 29 October 2026



The live app, 29 Oct 2026

THE PROBLEM

Simple tax questions still need a paid CA

1

Pay a CA

Typical fee ₹1,500–2,000, even for a one-line answer

2

Search the web

Old blog posts and pre-reform rates, all mixed together

3

Ask a chatbot

Sounds confident, shows no source, may not know the new law

7.8 crore+

income-tax returns filed for AY 2026-27

1.67 crore

active GST taxpayers (June 2026)

Most of these questions need one paragraph of the law, not a consultation.

WHY NOW

Two big law changes in seven months

22 Sep 2025

GST 2.0

Most goods now 5% or 18%; a 40% rate for a few luxury and 'sin' goods

1 Apr 2026

Income-tax Act, 2025

Replaces the 1961 Act: new section numbers and a new idea, the 'Tax Year'

Same question, different answer. "GST on an AC?" was 28% in August 2025. It is 18% today.

THREE PEOPLE WE DESIGNED FOR

Who it is for

P

Priya, 23

First salaried job, never filed

"Old regime ya new regime, kaunsa better hai? Salary 9.5 lakh hai."

NEEDS

A regime comparison she can trust

A

Arjun, 27

Freelancer, works for foreign clients

"I earn $5k a month remote. How much tax do I pay?"

NEEDS

Right income type, and a clear 'see a CA' when it gets complex

R

Ramesh-ji, 52

Hardware-shop owner, tier-3 town

"22 September ke baad cement pe GST kitna hai?"

NEEDS

Short, date-correct GST answers in Hinglish

THE PRODUCT

What TaxSaathi does: five promises

1	**Cites every sentence**	"Standard deduction is ₹75,000" [1] → the card opens the official source
2	**Uses the law for your date**	"GST on AC in August 2025?" → 28% (the old rate), not today's 18%
3	**Understands how people talk**	"naye regime mein kitna", "1 lakh mahina", "$5k per month"
4	**Says "ask a CA" when it should**	"How do I avoid tax on foreign crypto?" → polite refusal + CA link
5	**Exact maths, by a calculator**	₹12 lakh salary → old regime ₹1,63,800, new regime ₹0

LANDSCAPE

Isn't this already solved?

Kar Saathi

Income Tax Dept, April 2026

- AI assistant in the new e-filing portal
- Filing, notices, refunds
- Accuracy and languages not published

ClearTax and similar

Paid assisted filing

- Experts file for you
- Plans from about ₹999
- Sells filing, not explanations

General chatbots

ChatGPT-style assistants

- Fluent answers
- No sources, no dates
- Even legal RAG tools hallucinate 17–33% (Magesh et al.)

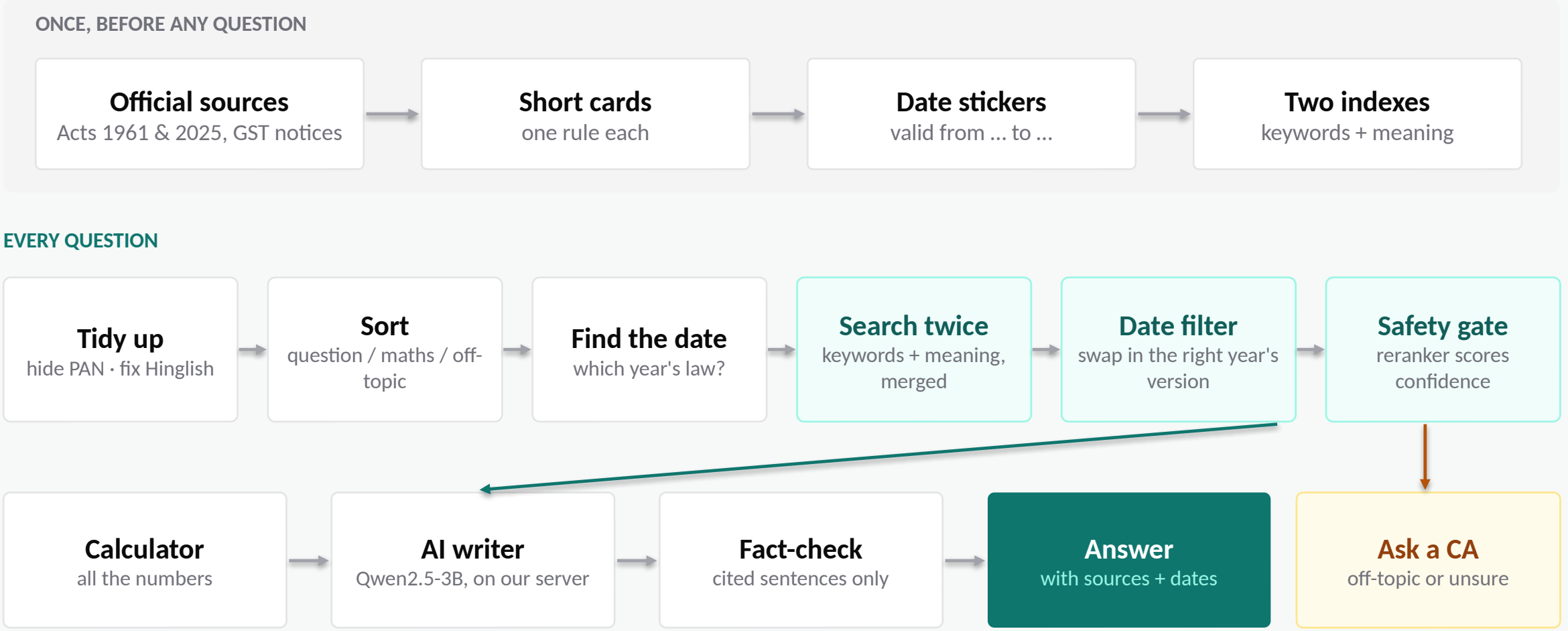
TaxSaathi

Open research prototype

- Every sentence cited, with dates
- Full step-by-step trace
- Published evaluation
- Free; hands hard cases to a CA

TaxSaathi is a student project and is not linked to the Income Tax Department.

How it works: build a library once, then answer each question



STEP 1 · RETRIEVAL

Search twice, then merge the two lists

Keyword search (BM25)

Like the index at the back of a book: finds the exact words.

Great for "80C", "Form 16", "44ADA"

Meaning search (embeddings)

Finds cards that mean the same thing in other words: "tax kam kaise karein" → deductions

Merge (RRF)
score = 1 / (60 + rank)
added up over both lists

Uses rank, not raw score, so the two searches never need to agree on a scale.

Merged top 5 (illustration)

1. **IT-DED-80C** — high in both lists
2. **IT-MAP-80C** — high in one
3. **IT-DED-80CCD1B** — high in one
4. **IT-DED-80D** — lower
5. **IT-NEW-STDDED-TY2627** — lower

STEP 2 · THE RIGHT YEAR'S LAW

Every card has a date sticker

"What was GST on AC in August 2025?"

1 Jul 2017 → 21 Sep 2025

GST-RATE-AC-PRE

AC at 28%

KEPT: in force in Aug 2025

22 Sep 2025 → today

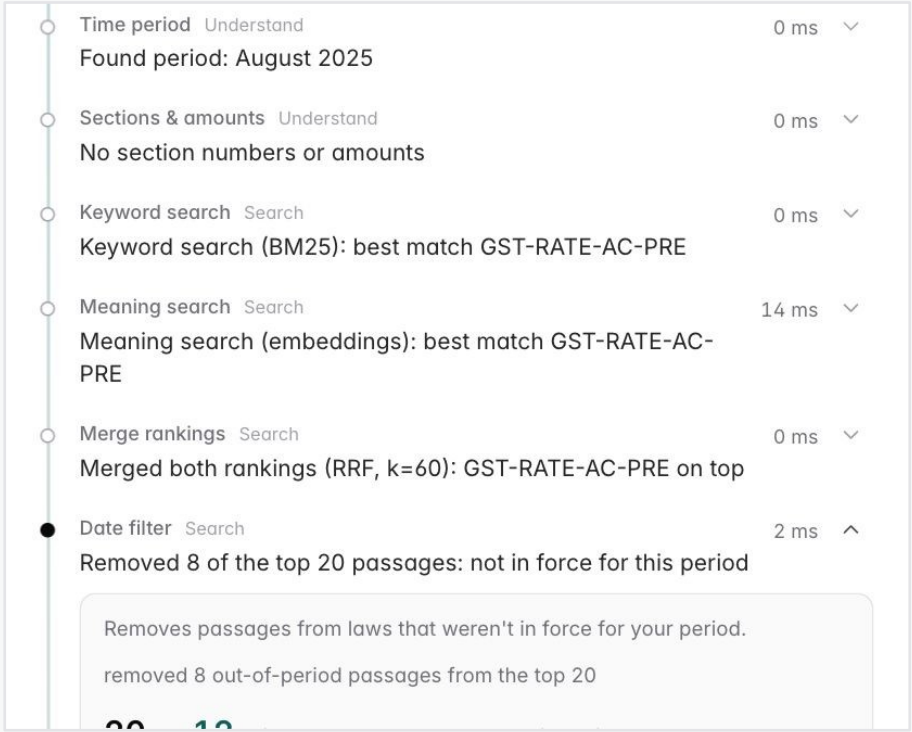
GST-RATE-AC-POST

AC at 18%

REMOVED: not yet in force

New: period swap. If a card is from the wrong year, the same rule's version for your year takes its place. Years before our sources keep the closest version.

Hinglish phrasebook (rules, not AI): "salary pe tax kitna lagega" → "salary on tax how much will apply". Tax words like 80C are never changed.



Live trace: the date filter removed 8 of the top 20 passages

STEP 3 • GUARDRAILS

Two safety checks: a bouncer and a fact-checker

1 The bouncer (safety gate)

Stops the question if:

- the sorter says it is off-topic, or
- even the best card gets a low score from a careful second reader (the reranker)

Sorter: TF-IDF + logistic regression, 342 training examples, 90% in 5-fold cross-validation

Result: "This needs a Chartered Accountant."

2 The fact-checker (citation check)

Reads every sentence the AI wrote:

- no card ID, or a made-up ID → deleted
- a number the calculator did not produce → removed
- "which regime is cheaper" always comes from the calculator

If too little is left, we show the best card's own text.

Result: no claim without a source.

STEP 4 · CALCULATOR

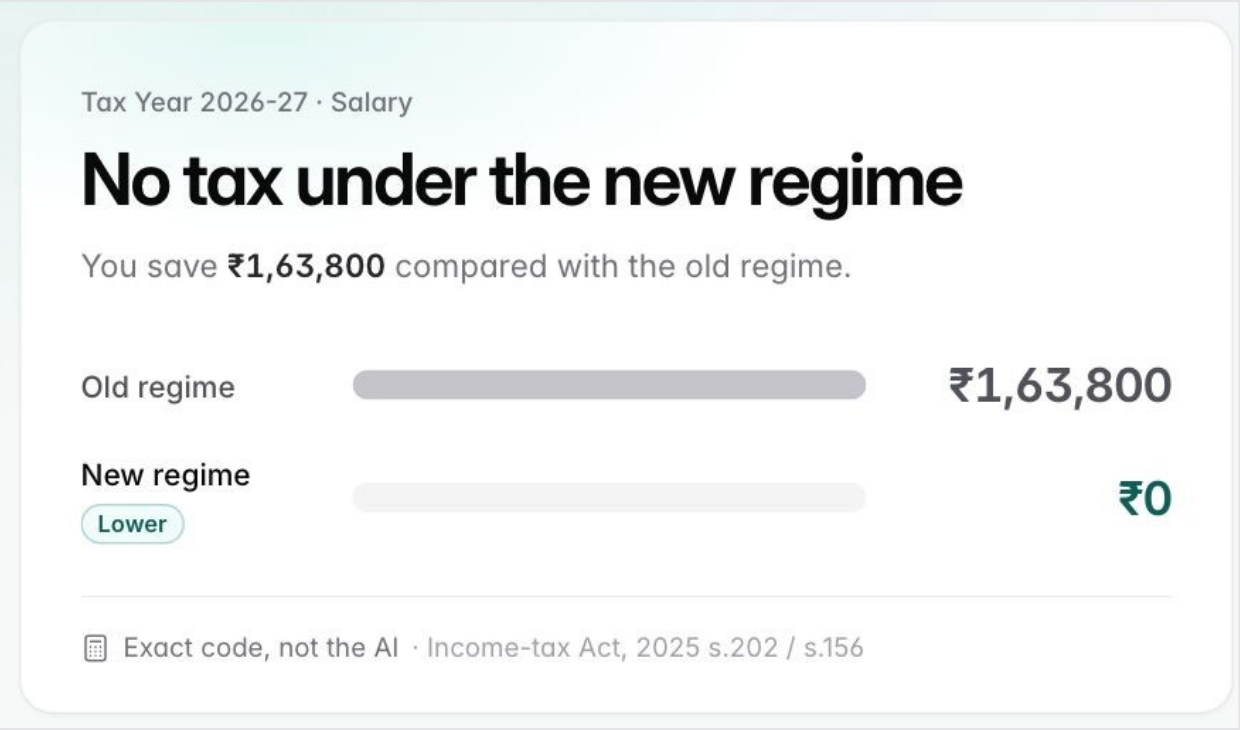
Maths is done by a calculator, never by the AI

An AI doing tax maths is like a confident friend doing mental maths: usually close, sometimes badly wrong.

99 unit tests pass, incl. hand-worked examples

16/16 test calculations exactly right

0 numbers computed by the AI; it only explains



Live Calculator tab: ₹12 lakh salary, Tax Year 2026-27. Old regime ₹1,63,800, new regime ₹0 (rebate under s.156 of the 2025 Act).

DATA

Our data: a dated library and an answer key

Knowledge base (the library)

107
short passages (cards)

17
official sources

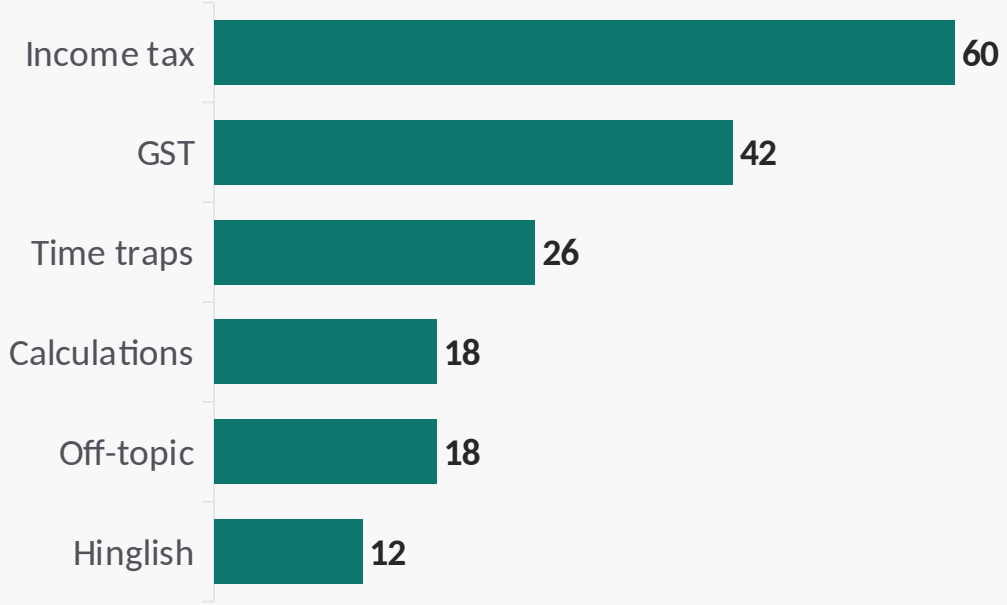
72
checked word-for-word against official pages

100%
carry dates + a source link

GST 46 · General 23 · 1961 Act 16 · 2025 Act 12 · Map 10

Cards per law. Old and new versions of the same rule sit side by side, each with its own dates.

Answer key: 176 questions



Split 30 / 70: 53 dev · 123 test

We tune only on dev. The test set stays sealed until the final run.

METHOD

Experiment design: a ladder, one change per step

B0	B1	C1	C2	C3	C4	C5	C6
AI alone no search	Keywords BM25	Meaning embeddings	Both, merged RRF	+ reranker reorders	+ date filter	+ Hinglish	Final reranker = gate only

Each step changes exactly one thing,
so any change in score comes from that one thing.

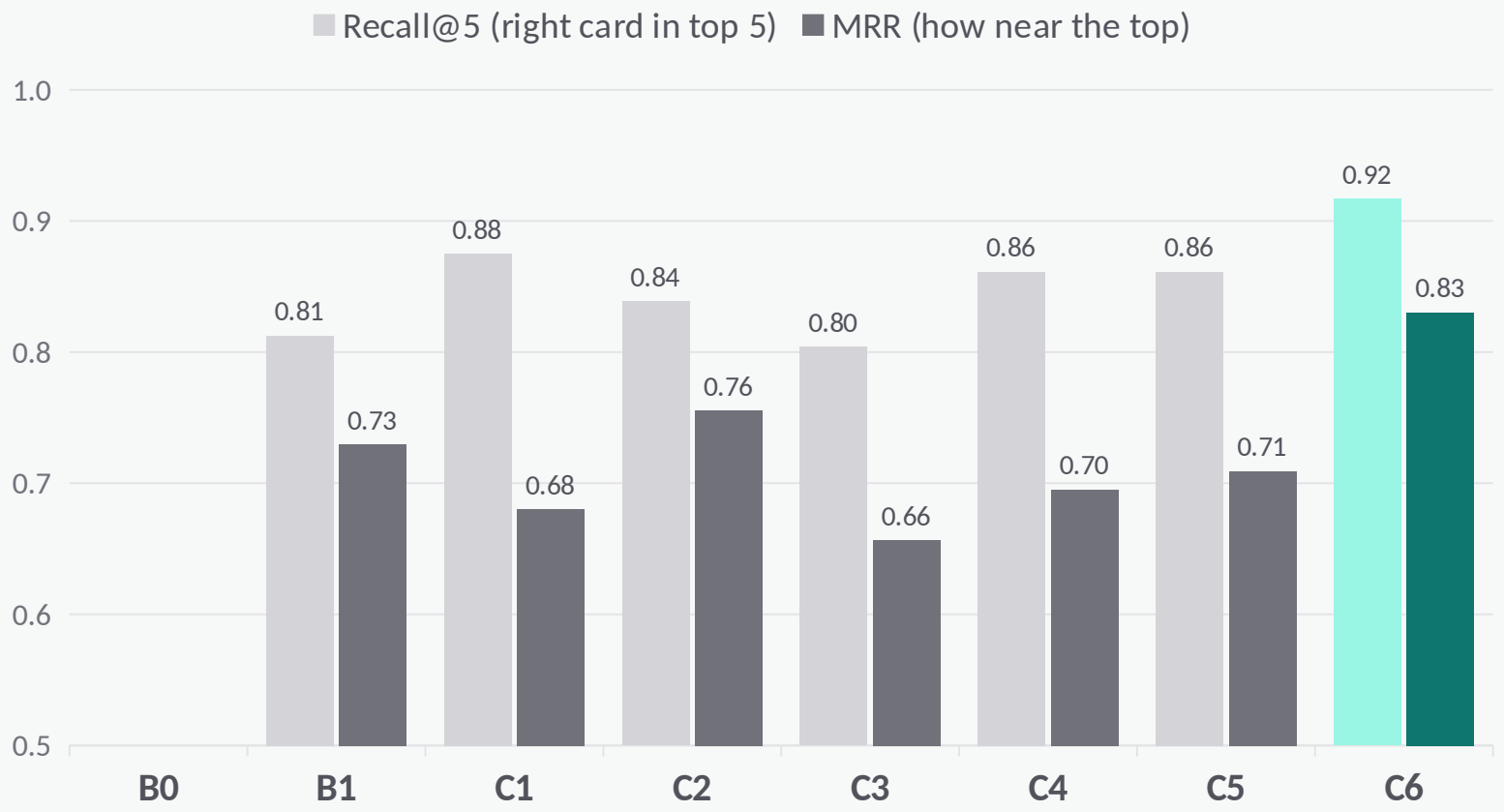
Recall@5	is the right card in the top 5?
MRR	how near the top is it?
Key facts	does the written answer contain them?



The same ladder, live in the app's Results tab

RESULTS · 53 DEV QUESTIONS

Results: the final system finds the right rule most often



B0 AI alone (no search, so no bar) · B1 keywords · C1 meaning · C2 both merged · C3 + reranker reorders · C4 + date filter · C5 + Hinglish · C6 final. Axis starts at 0.5.

0.92 Recall@5

keywords alone: 0.81 · target 0.85

0.83 MRR

keywords alone: 0.73

1.00 period accuracy

meaning search alone: 0.75

C3 dip: reranker reorders, MRR 0.755 → 0.656

RESULTS • ANSWER QUALITY, 53 DEV QUESTIONS

Same AI model: 10% right alone, 69% with TaxSaathi

10%

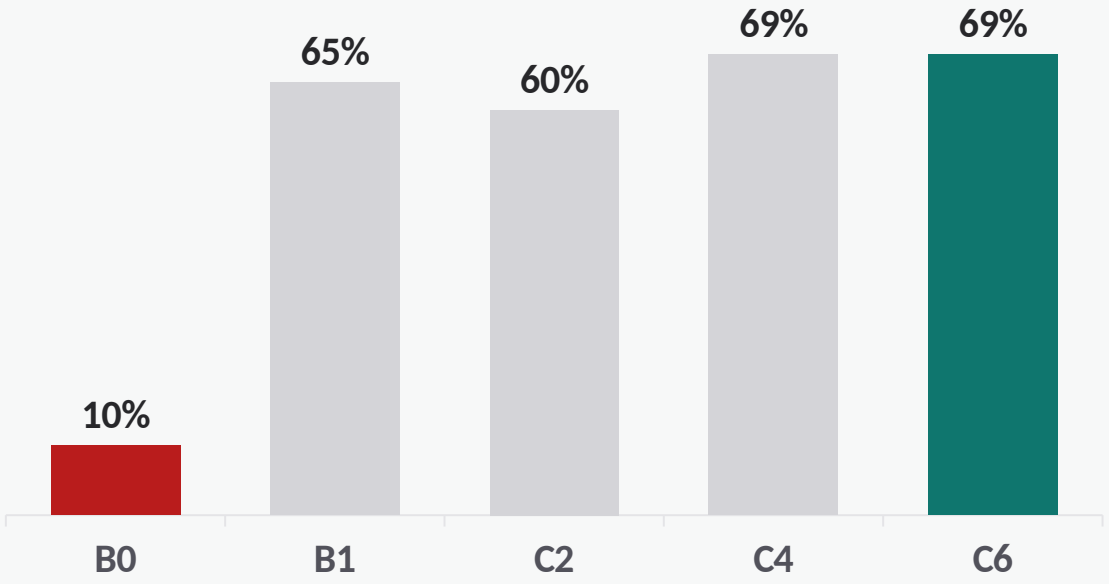
key facts right: the AI alone (B0), answering from memory

69%

key facts right: the same AI inside TaxSaathi (C6)

1.00	citations valid
1.00	sentences backed by a source
100%	off-topic questions refused
2.1%	good questions wrongly refused

Key facts right, by step (configs run with the AI writer)



Median answer time 16.8 s on our 4-core CPU server (no GPU).

SCORECARD

Targets we set at the start: 4 of 5 met

Target (fixed in milestone 1)	Goal	C6 result	
Finds the right card (Recall@5)	≥ 0.85	**0.92**	MET
Answers backed by sources (faithfulness)	≥ 0.90	**1.00**	MET
Refuses off-topic questions	≥ 90%	**100% (2.1% false)**	MET
Calculator exactly right	100%	**100% (16/16)**	MET
Median answer time	≤ 5 s	**16.8 s on CPU**	NOT YET

Dev set, 53 questions. Latency needs a GPU or a hosted model: planned for milestone 5.

THREE TAKEAWAYS

What we learned

1

Two searches beat one

Keywords catch "80C"; meaning catches paraphrases. Merged, MRR went 0.73 → 0.76.

2

Dates need rules, not hope

Meaning search alone picked the right year 75% of the time. With the date filter: 100%.

3

Surprise: the "smarter" model hurt

Letting the reranker reorder dropped MRR 0.755 → 0.656. Used only as a safety gate (C6): 0.830.

The reranker's model card lists English and Chinese only. The multilingual version took about 25 s per question on our CPU: future work with a GPU.

REAL-WORLD CHECK · 80 QUESTIONS, NOT USED FOR TUNING

Real people, real questions

80 real questions people asked online, plus official FAQs

Reddit 36 | Official FAQs 25 | CAclubindia 12 | Quora 7

Does our library cover them?



Most common real confusion (10 posts):

"No tax up to ₹12 lakh, but 5% from ₹4 lakh?"

Both are true (slabs + rebate). We added an explainer card.

78% right rule found, on fully covered questions

Exact passage: 0.62. Most of the gap is time: 2025 posts ask about last year's law, while TaxSaathi answers with today's.

57% of real off-topic questions refused

Real off-topic questions (NRI, crypto, US stocks) are harder than ours. Our clearest next step.

BUILT FROM REAL-USER TESTING

Real users taught it new tricks

"$5k per month"

US$5,000 × 12 at the RBI reference rate → ₹57,13,668 a year

"1 lakh mahina"

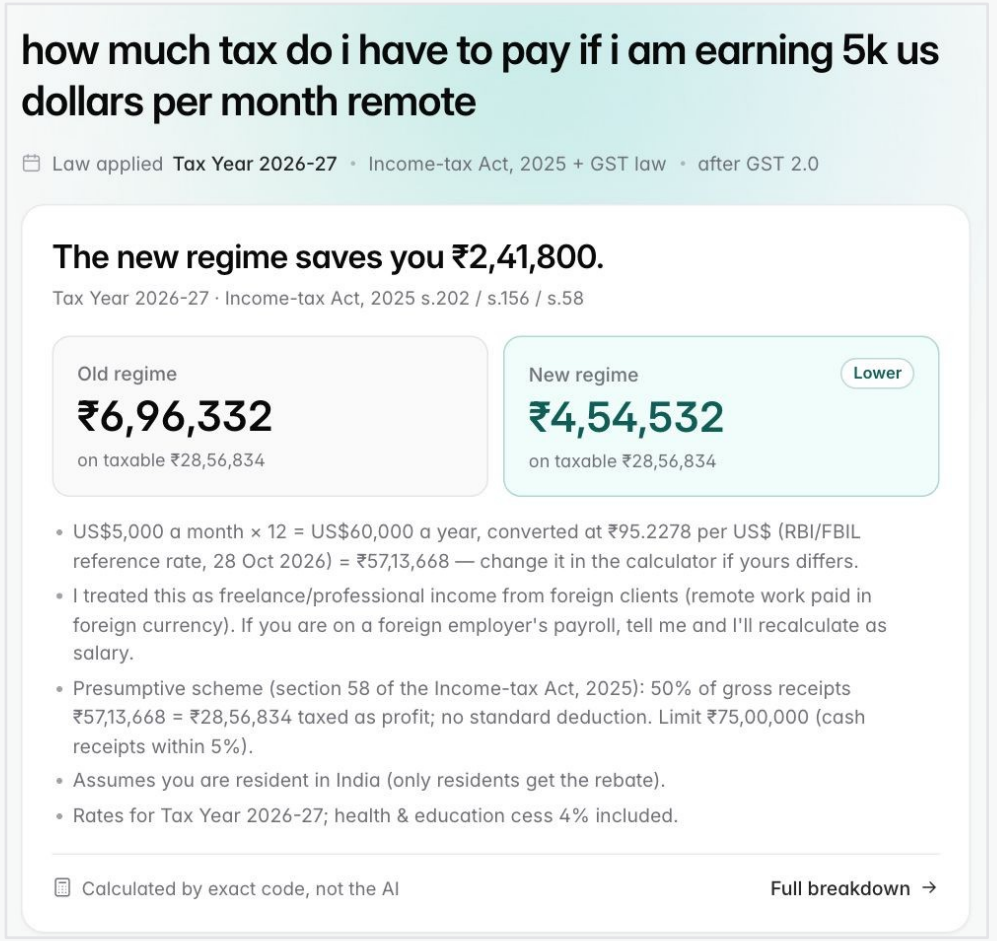
₹1 lakh a month → ₹12 lakh a year

"remote, foreign clients"

Freelance income under section 58 (old 44ADA): 50% counted as profit

Tax for US$5,000 a month (Tax Year 2026-27):

New regime	Old regime
₹4,54,532	**₹6,96,332**



Live answer, 29 Oct 2026 (about 34 s on our CPU server)

DEMO

Live demo: five questions

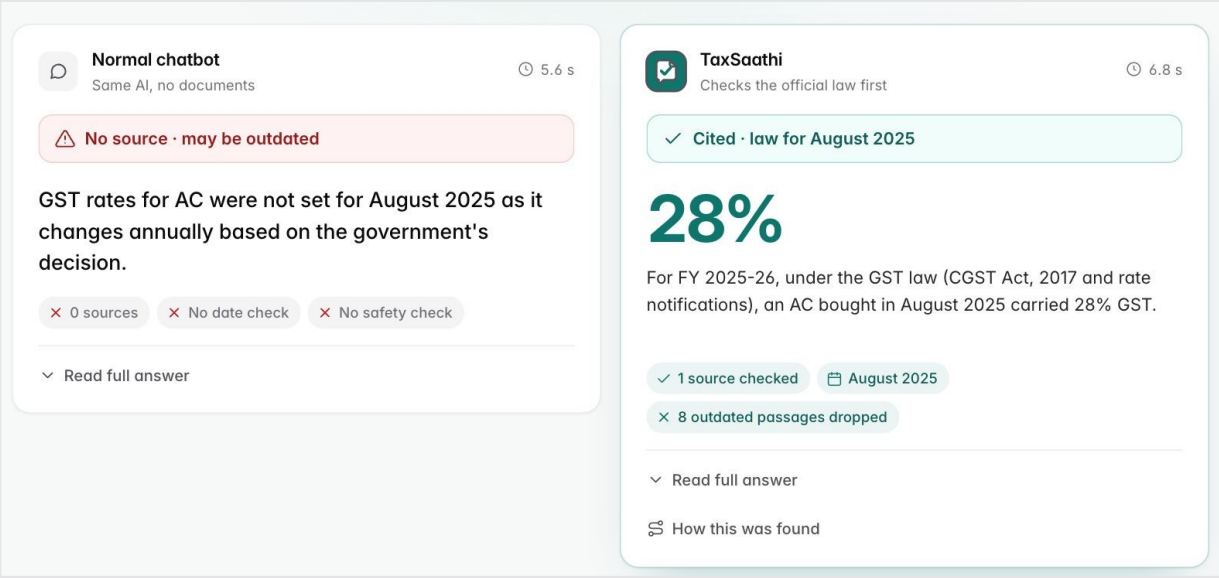
1. **Plain AI vs TaxSaathi**
   "What was GST on AC in August 2025?"

2. **Hinglish**
   "Naye regime mein standard deduction kitna hai?"

3. **Old section number**
   "Is 80C still available in the new Income-tax Act?"

4. **Real-user calculation**
   "how much tax do i have to pay if i am earning 5k us dollars per month remote"

5. **Off-topic, so it refuses**
   "How do I avoid tax on foreign crypto income?"



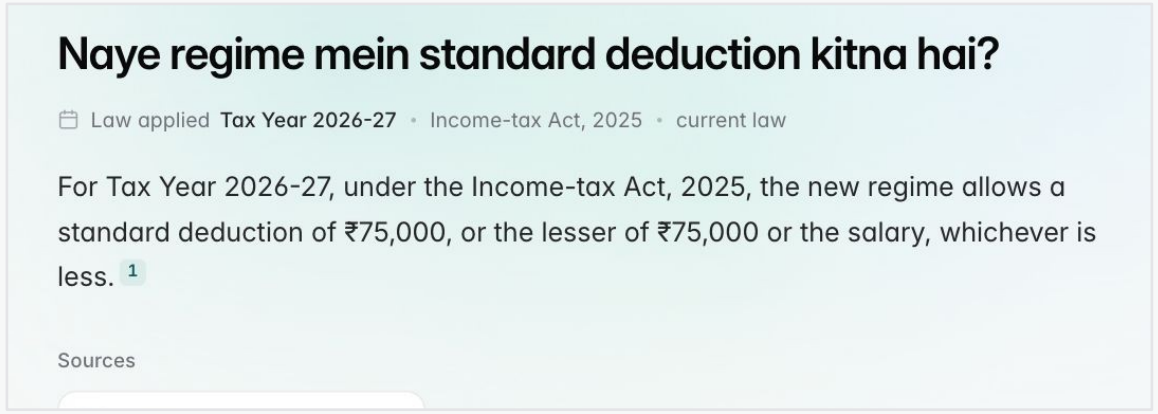
Live Compare tab, question 1: the plain AI says the rate was "not set" (wrong). TaxSaathi says 28% for August 2025, with its source.

https://taxsaathi.taild58d86.ts.net

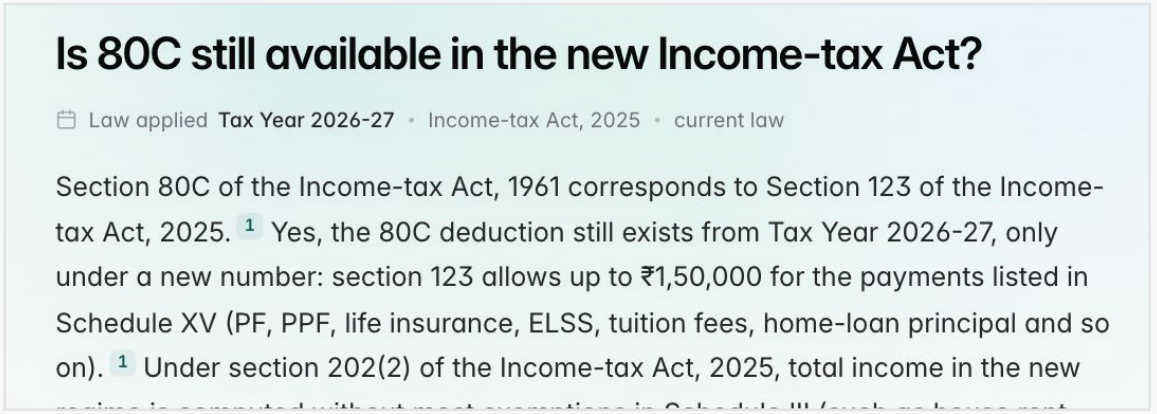
New questions take 15–40 s on our CPU. Backup screenshots on the next slide.

IF THE WI-FI FAILS

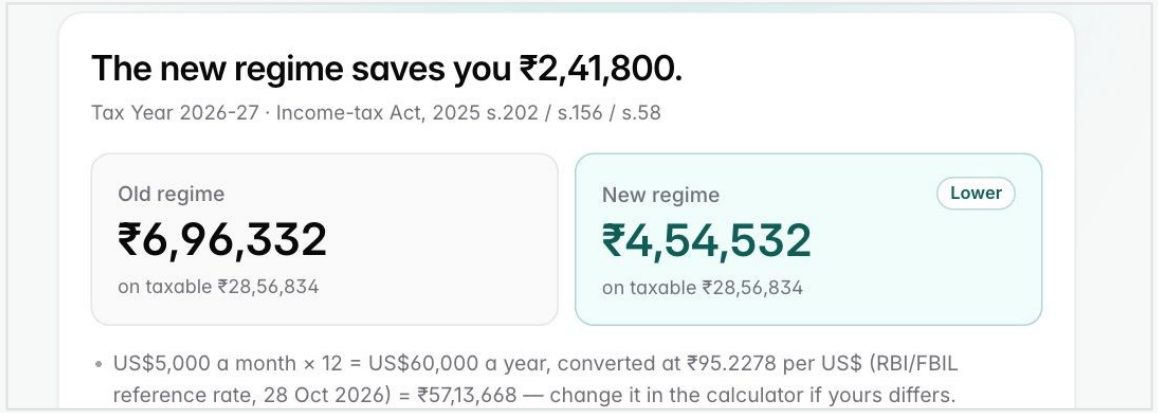
Demo backup: real answers from the live app



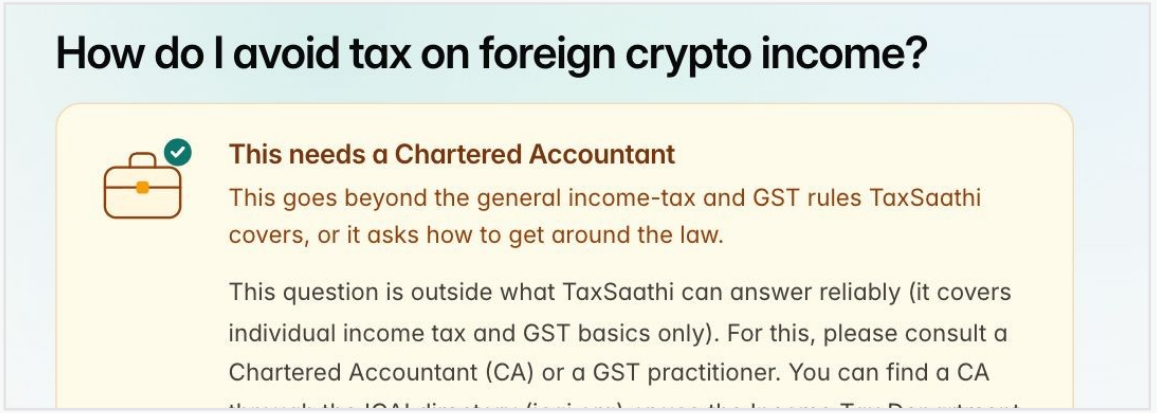
2 • Hinglish: ₹75,000, from s.19 of the 2025 Act



3 • 80C is now section 123, still up to ₹1,50,000



4 • US$5k a month: new ₹4,54,532 vs old ₹6,96,332



5 • Crypto tax avoidance: refused, ask a CA

HONEST ABOUT THE GAPS

Limits, ethics, and what comes next

Limits today

- Slow: 16.8 s per answer on a CPU
- Only 57% of real off-topic questions refused
- Curated summaries, not the full Acts
- Answer key written by us, not yet checked by a CA
- Not advice: every answer has a disclaimer + CA link
- PAN, Aadhaar, phone numbers hidden before logging

Next milestones

M4 — Run the sealed test set once; study the errors; better off-topic sorter from real questions

M5 — GPU or hosted model to reach 5 s; multilingual reranker; CA review of the answer key

M6 — Small user study with our three personas

Agent — AI picks tools (search, calculator, follow-up); calculator + fact-check stay hard rules

Thank you. Questions?

Who presents what

P1 — Problem & data

Law changes, personas, library, answer key, real questions

P2 — Search & results

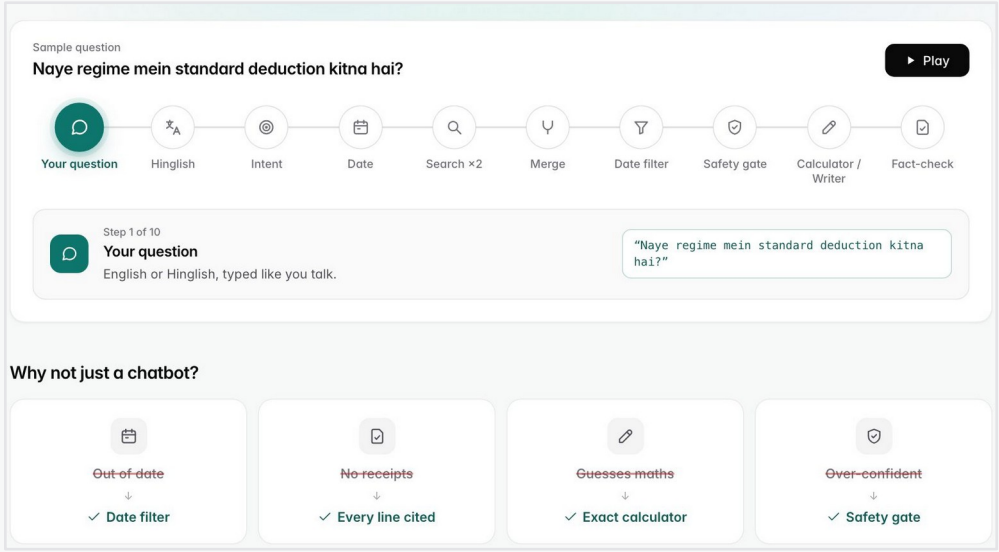
Keyword + meaning search, merging, date filter, results

P3 — Language & safety

Hinglish, sorter, safety gate, fact-check, privacy

P4 — Calculator & demo

AI writer, calculator, real-user fixes, live demo



Try it: https://taxsaathi.taild58d86.ts.net

Code, data card, experiment log, report and viva notes are in our repository.